Week 04 Progress Report

Details

- **Name of Student:** Tirthak Girish Likhar
- **PRN: 22070521041**
- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (26/01/2026 to 30/01/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of a Week

- During Week 4, I focused on understanding the foundational concepts of Security Operations Centers (SOC) and how security incidents are handled in real-world environments.

- My mentor guided me through SOC roles, the incident response lifecycle, and basic MITRE ATT&CK techniques.

- To connect theory with practice, I performed hands-on tasks to generate both normal and suspicious system activity and analyzed how these actions appear in Linux log files.

- This week helped me build a solid log analysis foundation, which is essential for the SIEM deployment and alert investigation work planned for the upcoming weeks.

Objectives of Week 4

- Learn SOC roles and responsibilities.
- Understand the SOC incident lifecycle.
- Study MITRE ATT&CK categories relevant to SOC detection.
- Observe how normal activity appears in system logs.
- Detect suspicious patterns without a SIEM.

Tasks Completed

1. SOC Fundamentals

- Studied SOC roles (L1, L2, L3).
- Understood the alert triage and escalation flow.
- Explored MITRE ATT&CK techniques.

2. Normal Log Generation & Analysis

- Generated login/session activity using whoami, who, and last.
- Created a normal file and observed file-related logs.
- Performed system update and package installation.
- Analyzed normal entries in syslog and auth.log.

3. Suspicious Log Simulation & Detection

- Attempted unauthorized access (cat /root) to create a suspicious log entry.
- Analyzed resulting authentication logs for abnormal activity.
- Used grep to filter failed or denied access attempts.

Key Accomplishments

- Developed a foundational understanding of SOC operations and attacker techniques.
- Improved skill in identifying and differentiating normal vs suspicious log activity.
- Learned how privileges, authentication events, and system operations appear in logs.
- Built confidence in log investigation techniques used by SOC analysts.
- Established a strong baseline for SIEM deployment in Week 5.

Tools & Commands Used

- `whoami`, `who`, `last`
- `touch`, `ls -l`
- `sudo apt update`, `apt install`
- `syslog`, `auth.log`
- `grep`, `tail`

Next Week Plan (Week 5)

- **Begin SIEM setup using Wazuh.**
- **Configure Wazuh server and web interface.**
- **Connect Ubuntu VM as an endpoint agent.**
- **Start collecting system logs inside SIEM.**
- **Validate initial ingestion and dashboard metrics.**

Screenshots of Activities (if Any)

Task 1 — SOC Fundamentals (Theory Section)

In this task, I learned the fundamental concepts of how a Security Operations Center (SOC) functions in real-world organizations. I studied the different SOC roles such as L1, L2, and L3 analysts, and understood how incidents move through the triage, investigation, containment, and recovery stages. This helped me understand what my responsibilities would be as an entry-level SOC analyst. My mentor also introduced me to the MITRE ATT&CK framework, which describes techniques attackers use during cyberattacks. This knowledge prepared me to analyze system logs in the next tasks because SOC analysts must know how attacker behavior appears in logs.

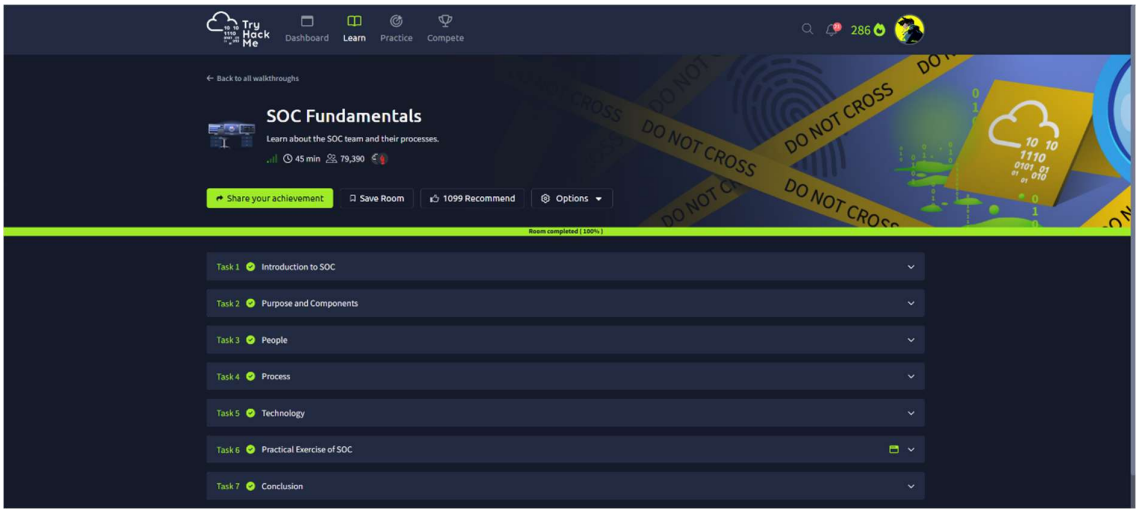

Fig 4.1.1: Completed Entire SOC Fundamentals Room on Tryhackme Hands-on-Platform as Instructed by industry mentor

Fig 4.1.2: Introduction to SOC (Security Operations Center)

Fig 4.1.3: The Main Purpose and Components of SOC team

Fig 4.1.4: The Three main Pillars of a SOC (People, Process, Technology)

Fig 4.1.5: Hierarchy of the entire Security Operations Department

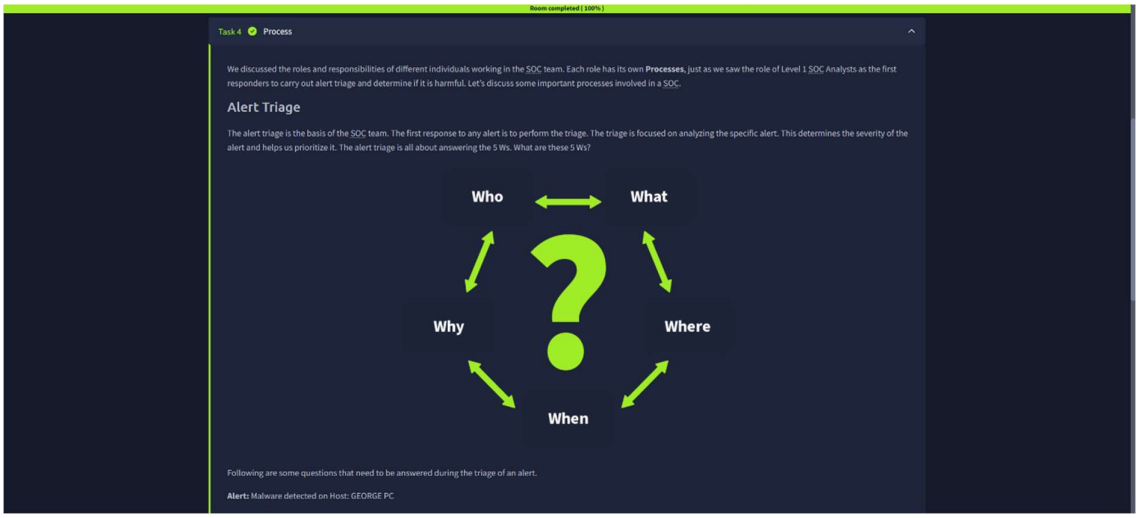

Fig 4.1.6: The Process (Alert Triage)

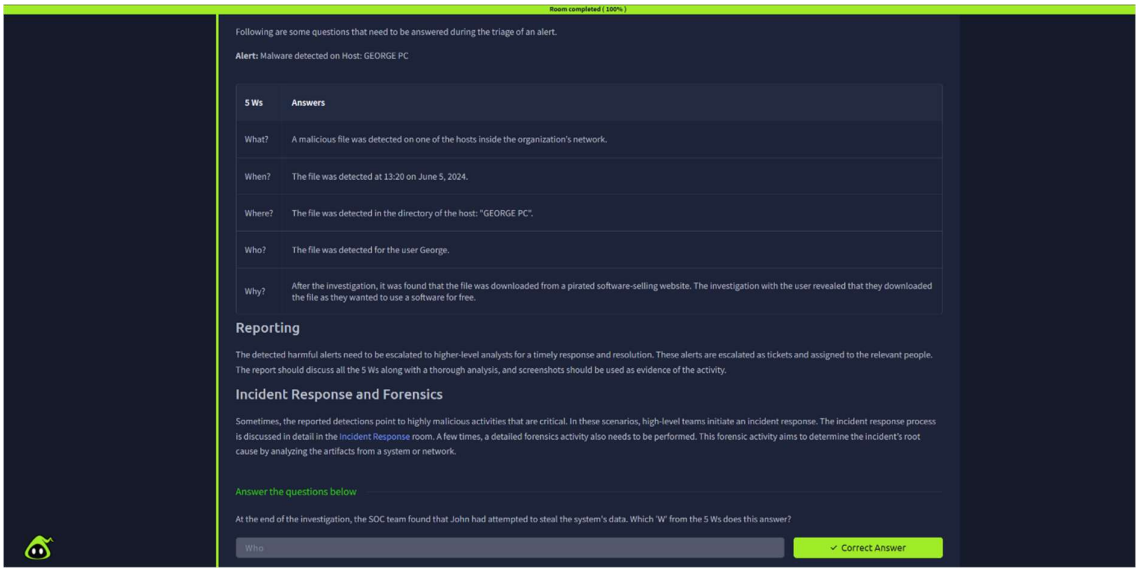

Fig 4.1.7: Other important aspects of SOC such as Reporting and Incident Response and Forensics

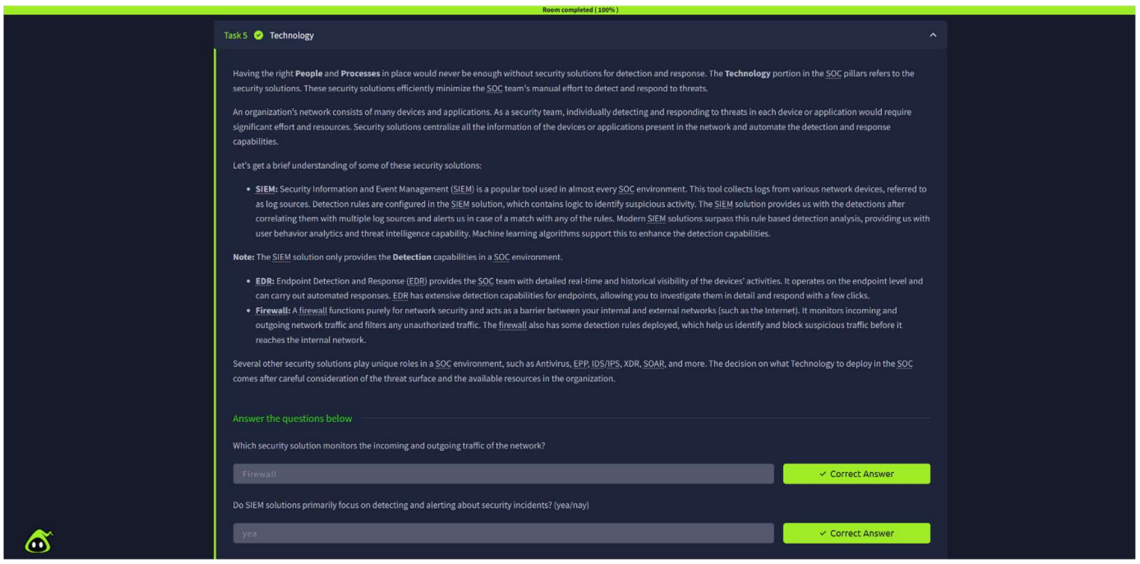

Fig 4.1.8: Important Technology and Security solutions such as (EDR, XDR, Firewall, SIEM)

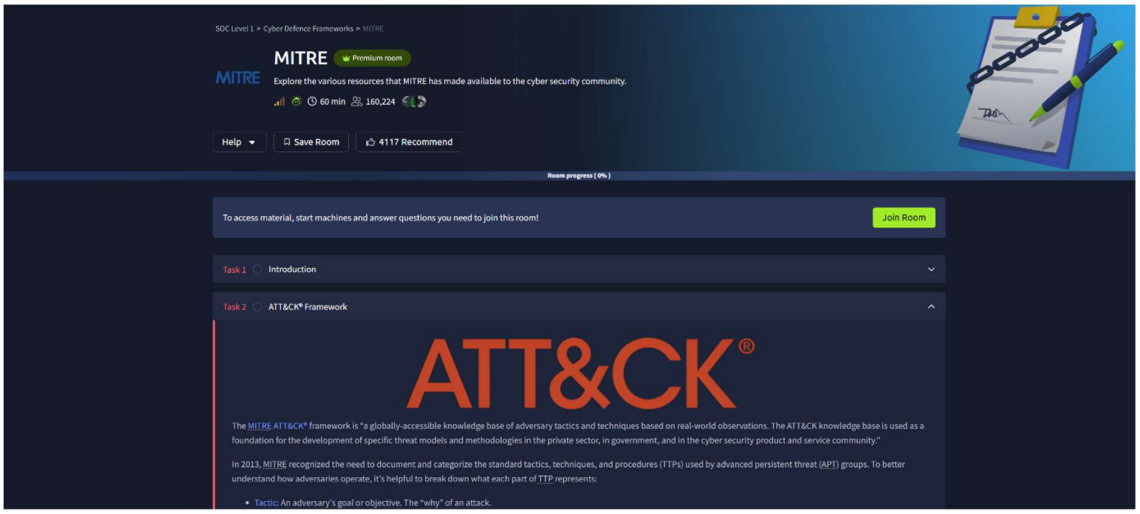

Fig 4.1.9: MITRE - ATT&CK framework Hands-on-Room on Tryhackme

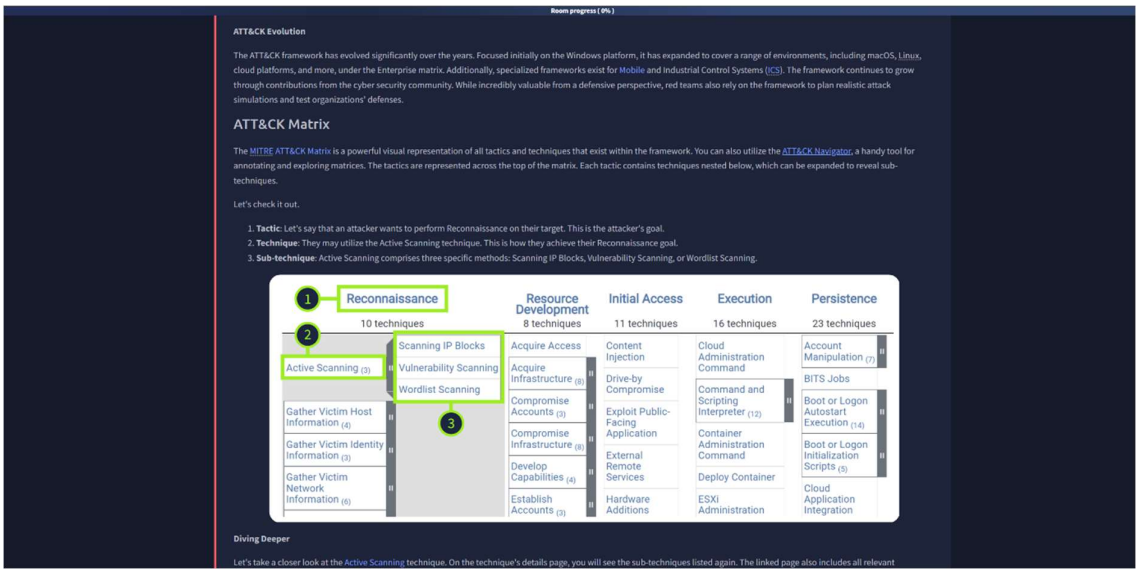

Fig 4.1.10: Attack Evolution

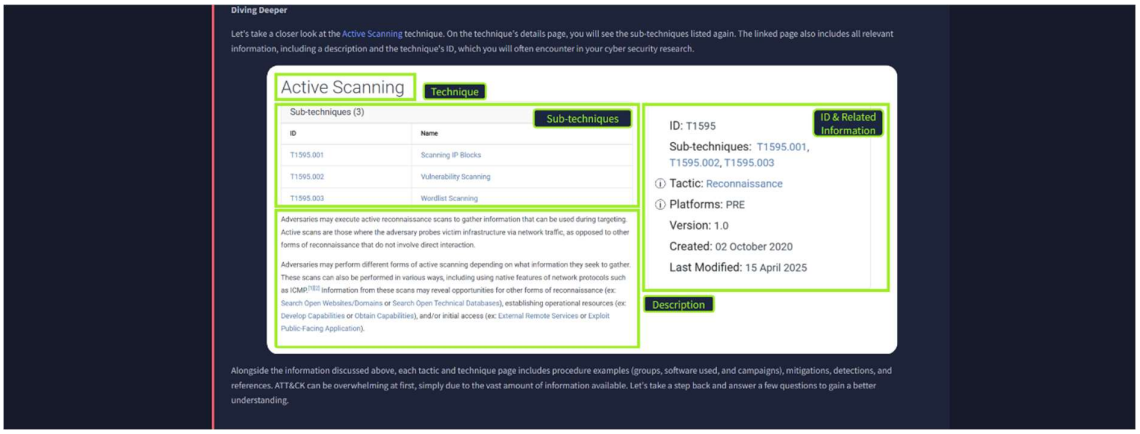

Fig 4.1.11: MITRE ATT&CK matrix

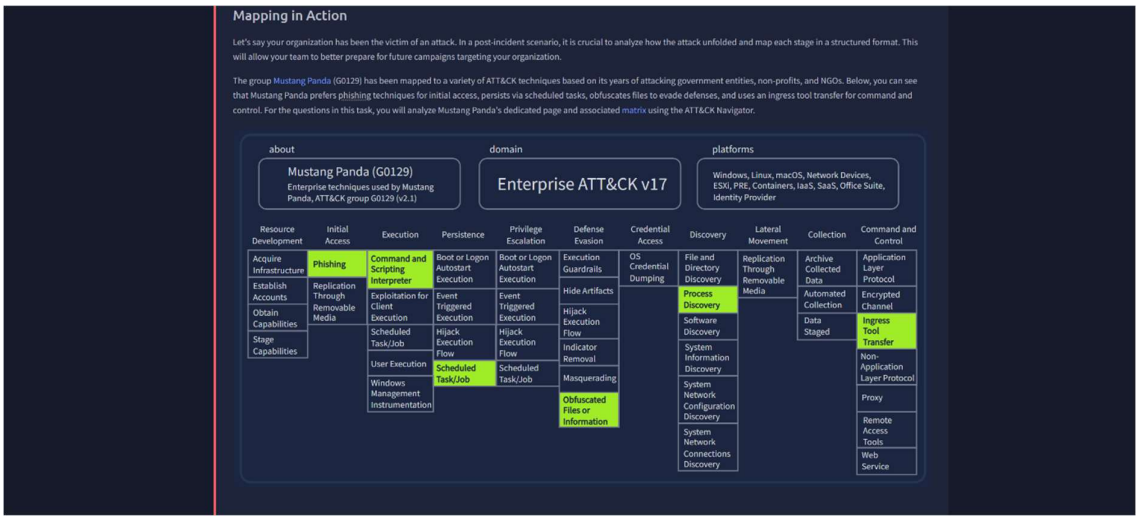

Fig 4.1.12: Enterprise Mapping in Action

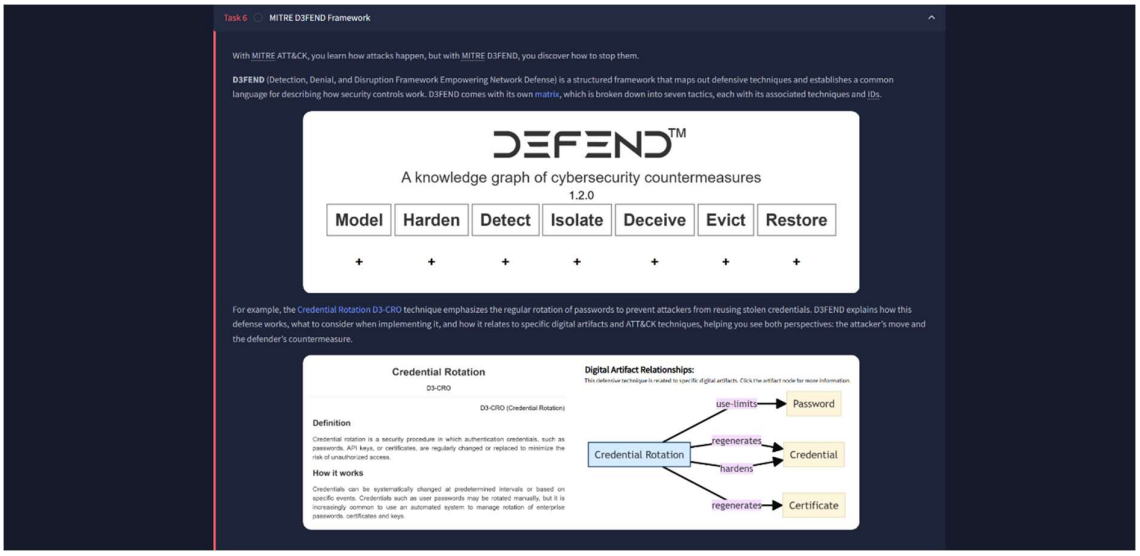

Fig 4.1.13: MITRE D3FEND Framework

Task 2 — Normal Log Generation & Observation

In this task, my goal was to generate normal system activity and observe how these events are recorded inside various Linux log files. This helps establish a "baseline," which means understanding how normal system behavior looks so that suspicious activities can be detected easily in the future. I performed multiple actions such as checking login history, creating a test file, updating packages using apt, and running sudo commands. Each of these actions creates entries in logs like syslog and auth.log. By observing these logs, I learned how routine user activity, system updates, and authentication events get tracked. This hands-on exercise helped me understand what normal logs look like before I move to SIEM setup next week.

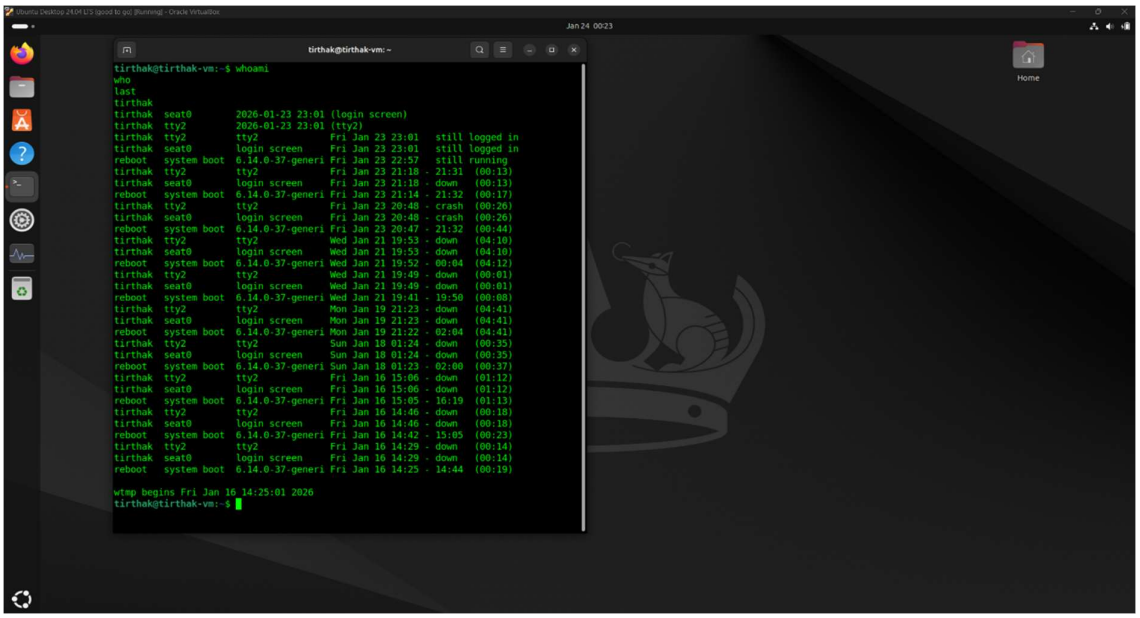

Fig 4.2.1 – Login and session logs using whoami, who, and last .

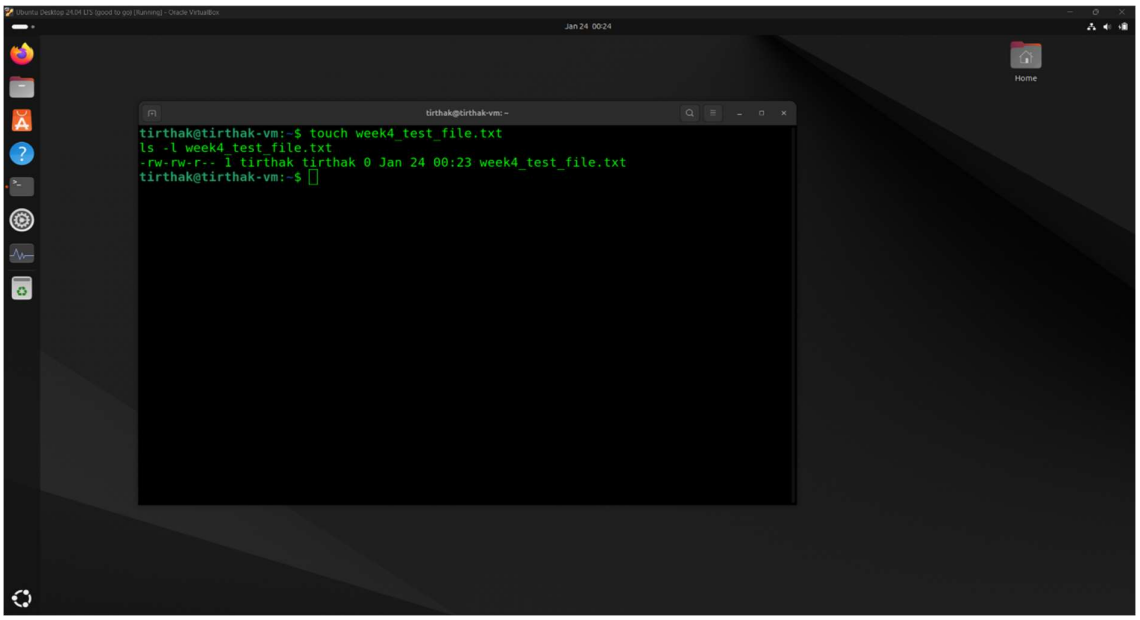

Fig 4.2.2 – File creation event using touch and file listing using ls -l.

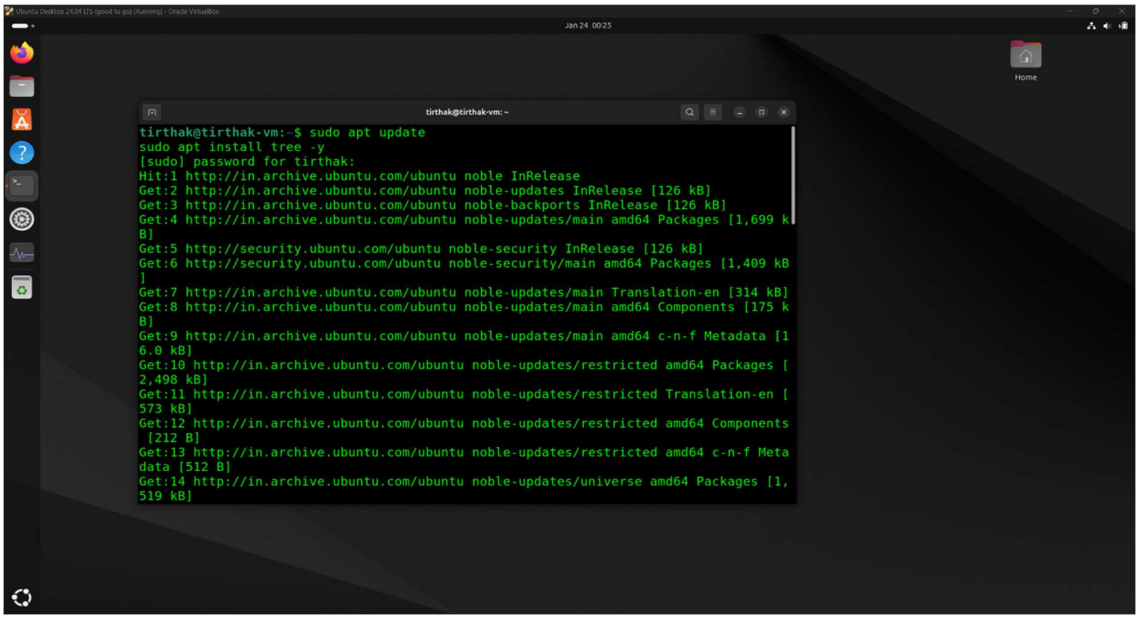

Fig 4.2.3 – System update and package installation logs from apt update and apt install.

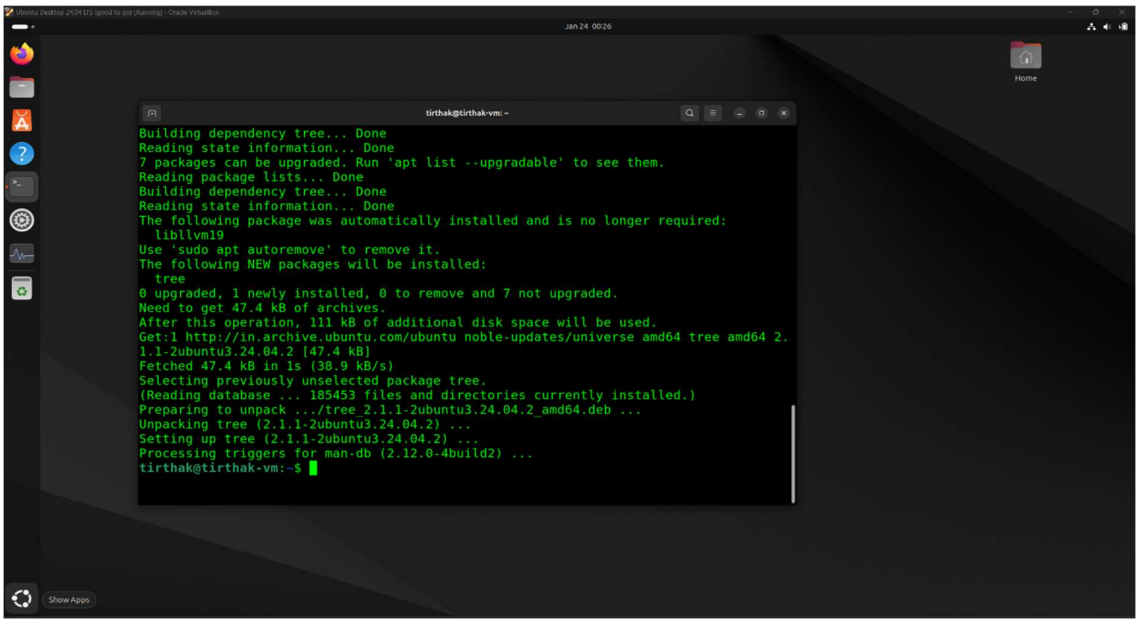

Fig 4.2.3 – System update and package installation logs from apt update and apt install.

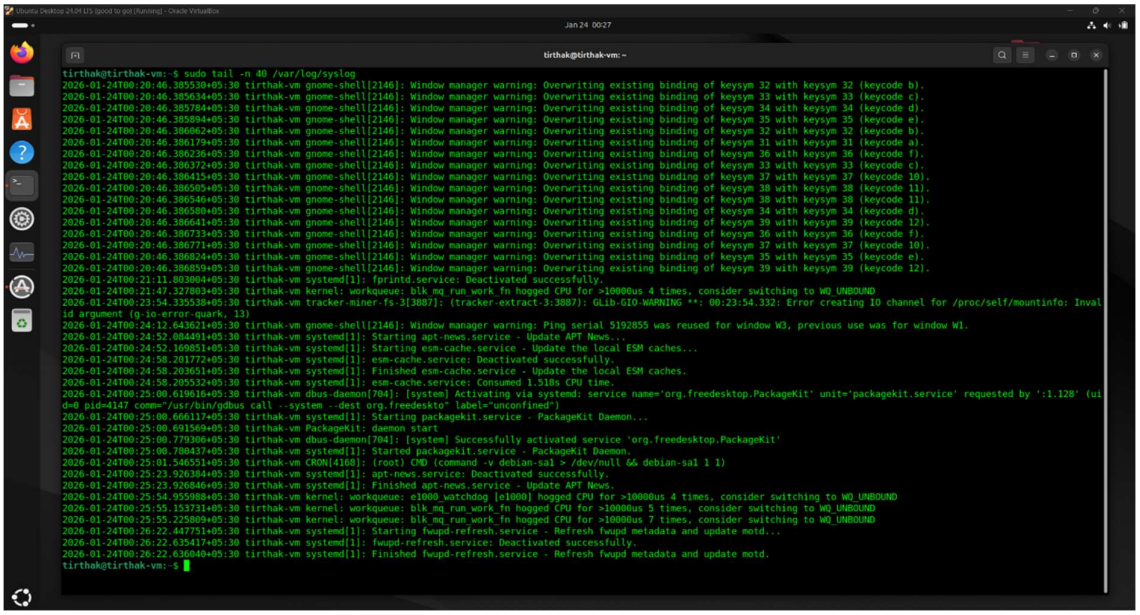

Fig 4.2.4 – Recent normal system events viewed from /var/log/syslog.

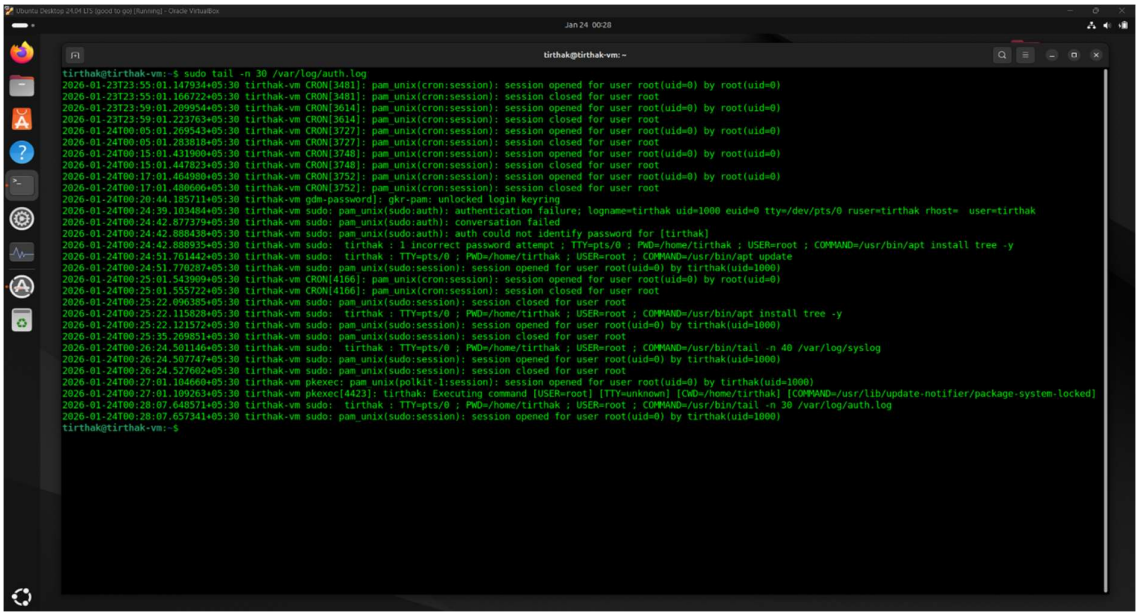

Fig 4.2.5 – Authentication and sudo activity logs from /var/log/auth.log.

Task 3 — Suspicious Log Pattern Detection

In this task, I simulated a harmless suspicious activity to understand how abnormal events appear in Linux logs. SOC analysts must be able to differentiate between normal behavior and actions that may indicate unauthorized access, privilege escalation attempts, or malicious intent. To practice this, I intentionally tried to access a restricted directory (/root) without using sudo. This action gets recorded as a permission-denied or unauthorized attempt. After performing this action, I checked the authentication logs to locate the suspicious entry. I also used filtering commands like grep to identify failed or denied attempts. This activity helped me understand how suspicious patterns look in logs and prepared me for detecting real alerts once I set up the SIEM in Week 5.

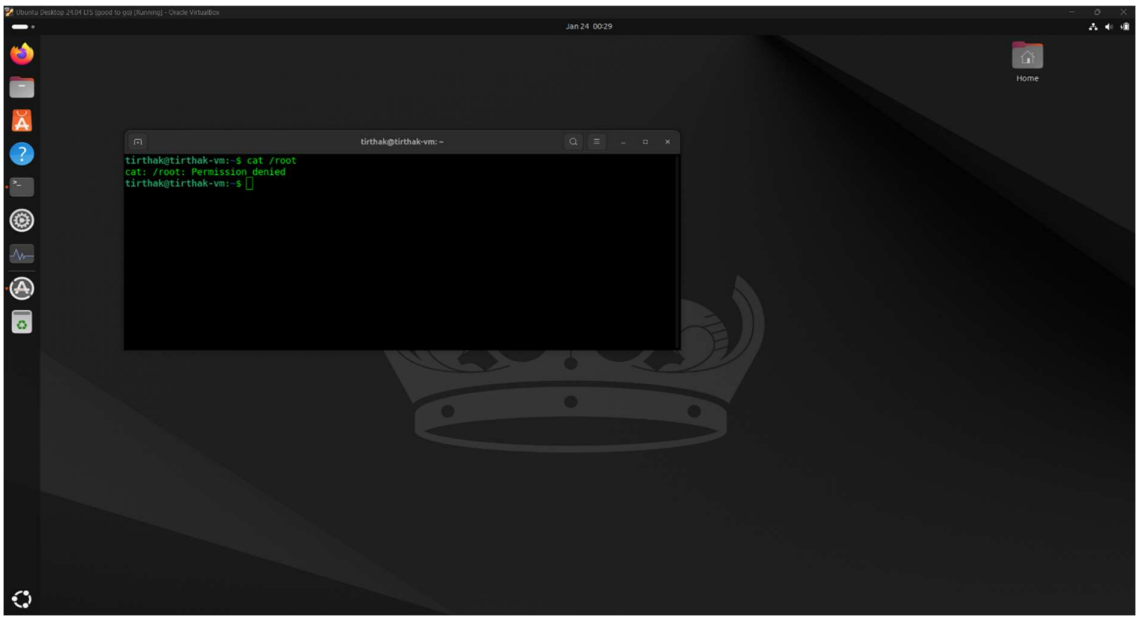

Figure 4.3.1 – Unauthorized access attempt to /root without sudo (permission denied event).

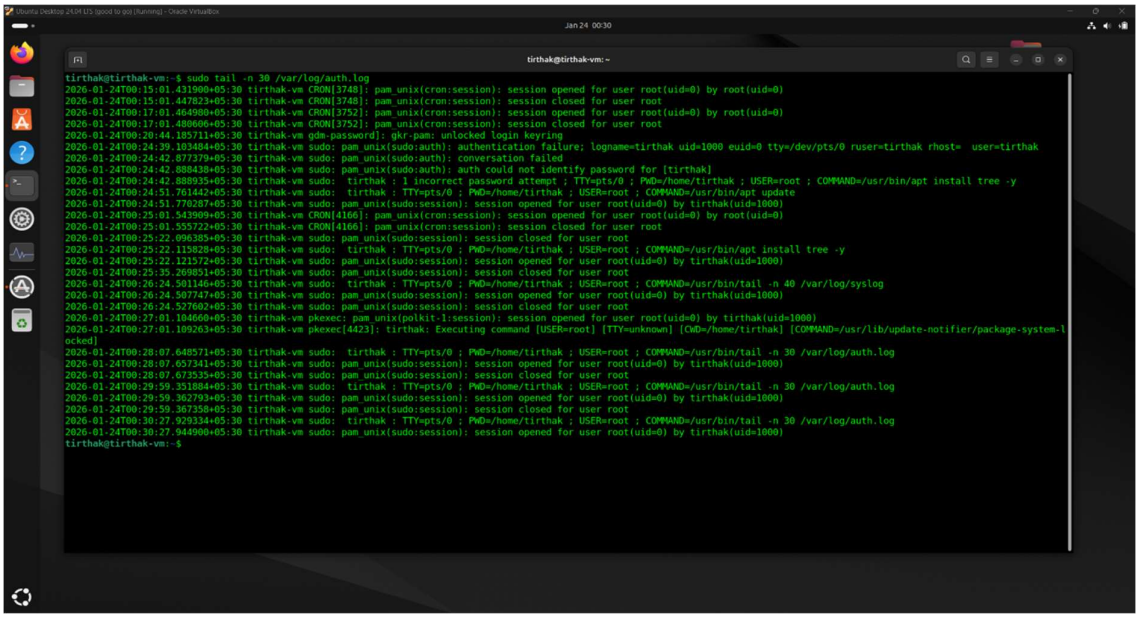

Fig 4.3.2 – Suspicious authentication event identified in /var/log/auth.log.

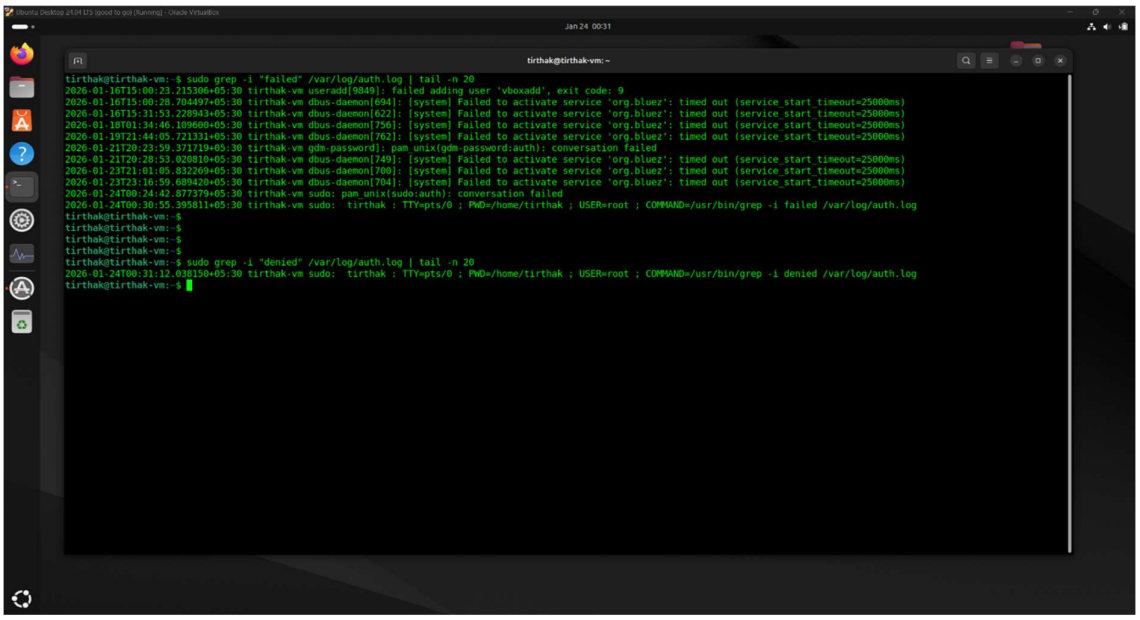

Figure 4.3.3 – Filtered suspicious log entries using grep for failed/denied attempts.